



Simulation-based inference of the 3D shapes of central galaxies in halos: preparation for intrinsic alignment studies

Antonin Corinaldi, PhD student at CEA Paris-Saclay

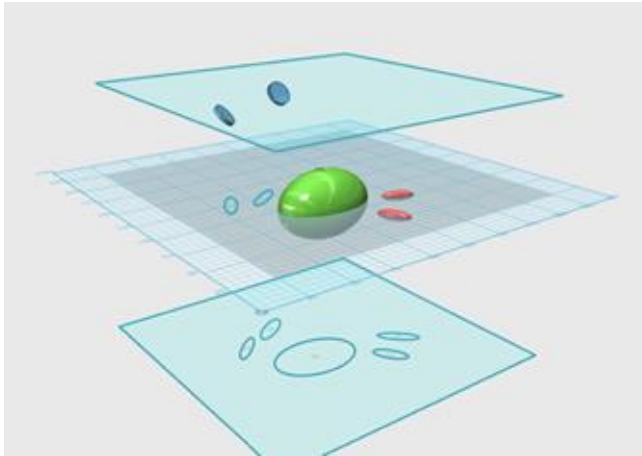
Supervisors: Martin Kilbinger, Sandrine Codis, Calum Murray,
Romain Paviot

EchoIA meeting - September 3, 2026

Context and motivations

IA with galaxy 3D morphology and galaxy-halo connection

- The morphology of galaxies in 3D encodes fundamental informations about processes governing galaxy formation and evolution within the cosmic web.



Joachimi et al. 2026



Fortuna & Chisari (2025)

- This is interesting for IA: the IA signal depends on the morphology of galaxies and the way their shapes evolved in 3D inside halos among the large-scale structures of the Universe

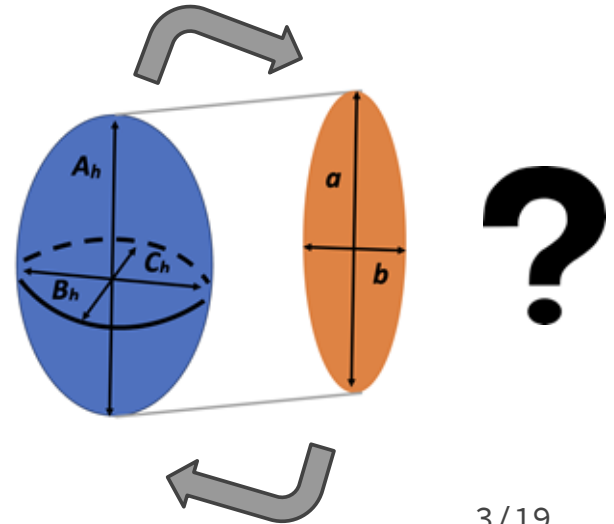
Objectives

- **Aims:**

- 1) Developing a forward model for the 3D morphology of central galaxies inside dark matter halos using N-body simulations
- 1) Inferring the galaxy-halo connection and the distribution of the shapes of galaxies in 3D from their observed projected images

- **Problem:**

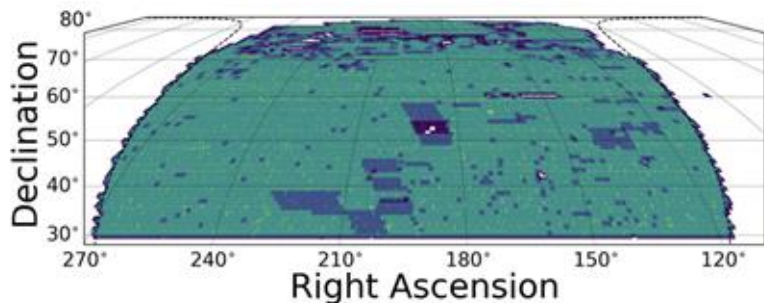
=> *How to go from projected images to 3D shapes ?*



UNIONS (Ultraviolet Near-Infrared Optical Northern Survey)



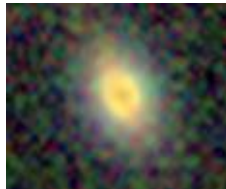
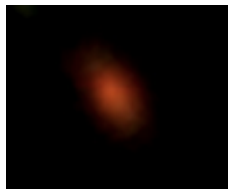
- CFHT, Pan-STARRS, Subaru Telescope (u,g,r,i,z-band)



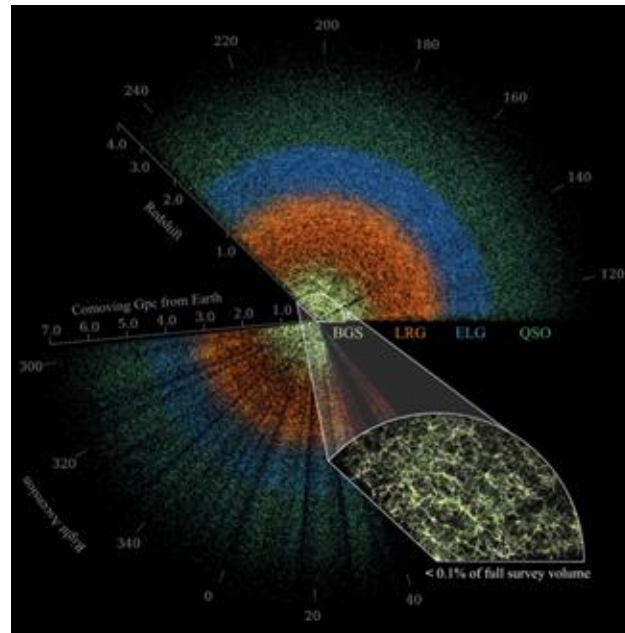
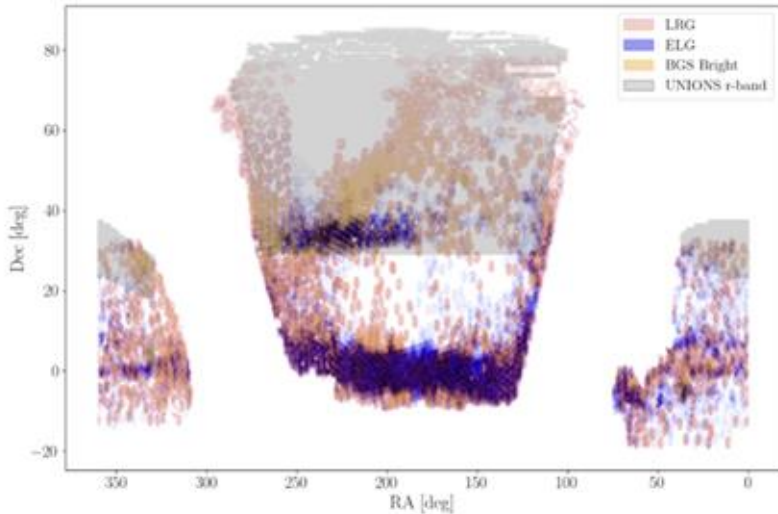
- Photometric galaxy survey in the northern hemisphere
~340 millions of galaxy shapes (r-band)
Footprint: 5800 deg² ([Gwyn et al. 2025](#))
- Measurement of intrinsic alignments by cross-matching UNIONS shapes of galaxies with spectroscopic surveys ([Hervas Peters et al. 2025](#))

UNIONS x DESI

- Spectroscopic samples of galaxies:
ELG, **LRG** and **BGS Bright** (DESI DR1)



Credit: DESI Legacy Imaging Survey

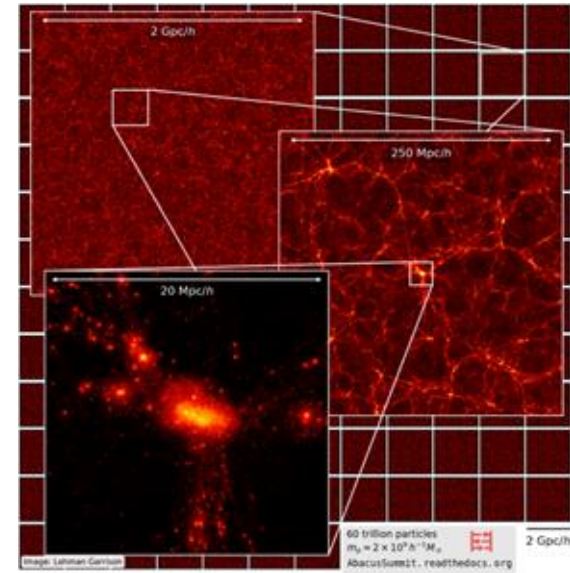
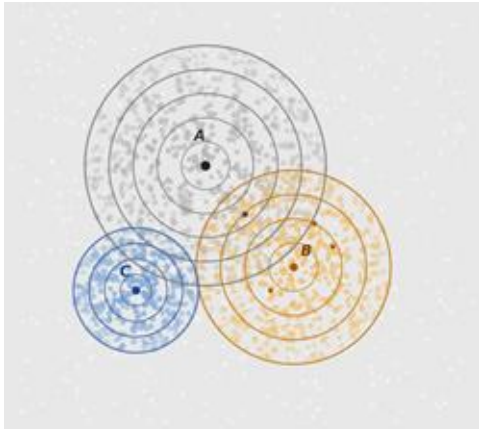


DESI Collaboration 2025

- Cross-match with **UNIONS** shapes
 - **ELG-UNIONS**: ~360 000 galaxies
 - **LRG-UNIONS**: ~360 000 galaxies
 - **BGS Bright-UNIONS**: ~1 300 000 galaxies

N-body simulations AbacusSummit

- DM-only simulation suite designed to meet the requirements of DESI
- Simulation box of size $2 h^{-1}\text{Gpc}$ at different snapshots
- Planck 2018 cosmology, 330 billion dark matter particles of mass $2 \times 10^9 h^{-1} M_{\odot}$



Maksimova et al. 2021

- Catalogues of halos identified with the CompaSO (Competitive Assignment to Spherical Overdensities) algorithm (*Hadzhiyska et al. 2021*)

Methodology

I) Getting a halo catalogue per galaxy sample

II) Modelling the 3D shapes of halos

III) Modelling the 3D shapes of central galaxies in their host halos

IV) Inferring galaxy-halo connection from observations using simulation-based inference (SBI)

I) Getting a halo catalogue per galaxy sample

- Step 1: generate a **mock galaxy catalogue** in a simulation box using a **HOD model** for each galaxy sample

Example: HOD model for the LRG ([Yuan et al. 2023](#))

Mean occupation number of **central** and **satellite** galaxies:

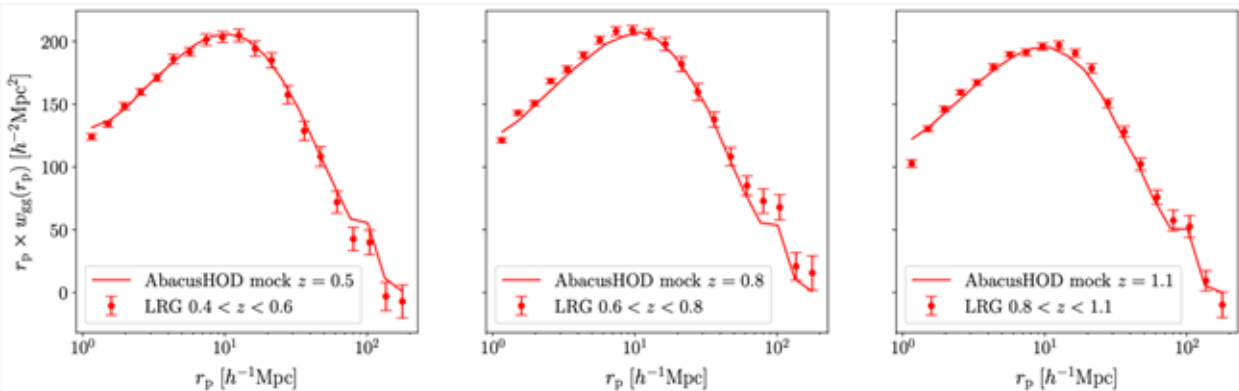
$$\bar{n}_{\text{cent}}^{\text{LRG}}(M) = \frac{f_{\text{ic}}}{2} \text{erfc} \left[\frac{\log_{10}(M_{\text{cut}}/M)}{\sqrt{2}\sigma} \right] \quad \bar{n}_{\text{sat}}^{\text{LRG}}(M) = \left[\frac{M - \kappa M_{\text{cut}}}{M_1} \right]^{\alpha} \bar{n}_{\text{cent}}^{\text{LRG}}(M)$$

M	mass of the host halo
M_{cut}	minimum halo mass to set a central galaxy
f_{ic}	incompleteness parameter (by definition: $0 < f_{\text{ic}} \leq 1$)
σ	steepness of the transition from 0 to 1 in the number of central
κM_{cut}	minimum halo mass to host a satellite galaxy
α	power law index on the number of satellite galaxies
M_1	typical halo mass that hosts one satellite galaxy

Code: *AbacusHOD* ([Yuan et al. 2021](#)), scripts here:  [antonincorinaldi](#)

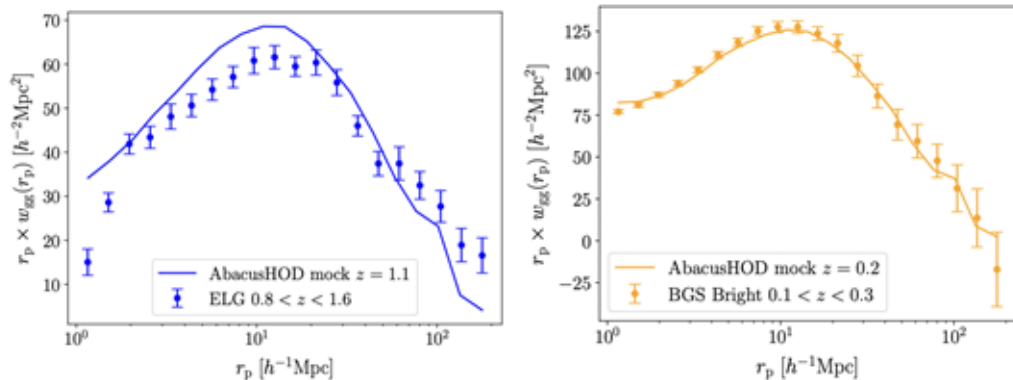
I) Getting a halo catalogue per galaxy sample

- Step 2: **clustering measurements** with the data and the mock and check the consistency between both (*Landy & Szalay 1993*)



$$\xi_{gg}(r_p, \Pi) = \frac{DD(r_p, \Pi) - 2 DR(r_p, \Pi) + RR(r_p, \Pi)}{RR(r_p, \Pi)}$$

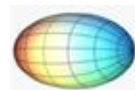
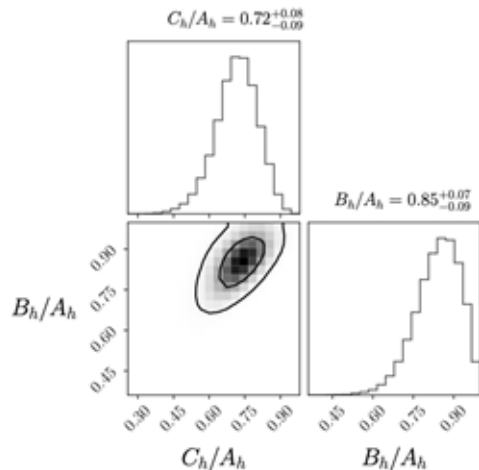
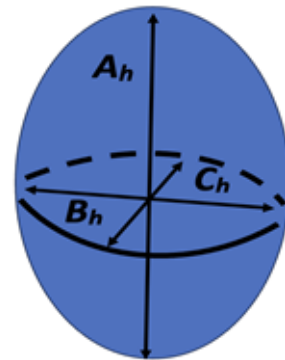
$$w_{gg}(r_p) = \int_{-\Pi_{\max}}^{\Pi_{\max}} d\Pi \xi_{gg}(r_p, \Pi)$$



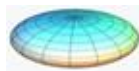
- Best fit HOD parameters:
 - LRG *Yuan et al. 2023*
 - ELG *Rocher et al. 2024*
 - BGS *Prada et al. 2025*

II) Modelling the 3D shapes of halos

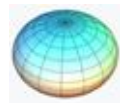
- Simple inertia tensor:
$$I_{ij} = \frac{\sum_n m_n x_{ni} x_{nj}}{\sum_n m_n}$$
- Axis lengths: square root of the eigenvalues of I_{ij} :
$$A_h > B_h > C_h$$
- 3D orientation: eigenvectors of I_{ij}



prolate $A_h > B_h \approx C_h$



oblate $A_h \approx B_h > C_h$



spheroidal $A_h \approx B_h \approx C_h$

Example: Abacus halos at $z = 0.5$

Pandya et al. 2024

III) Modelling the 3D shapes of central galaxies in their host halos

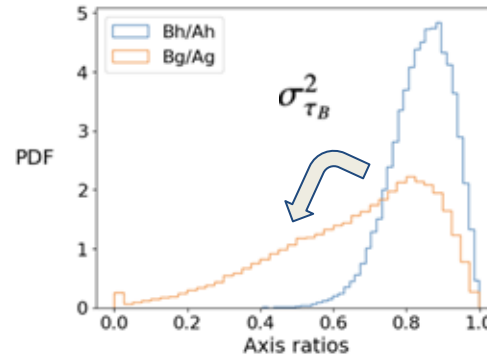
- Populating each halo with a 3D central galaxy with axis lengths A_g, B_g, C_g such that:

$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h} \quad \frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h} \quad \text{with } \tau_B, \tau_C \in [0; \tau_{\max}]$$



- Major axis of the central galaxy aligned with the host halo one
- 3D population of galaxies = gaussian draw of τ_B, τ_C around mean values $\mu_{\tau_B}, \mu_{\tau_C}$ and with a covariance matrix Σ :

$$\Sigma = \begin{pmatrix} \sigma_{\tau_B}^2 & 0 \\ 0 & \sigma_{\tau_C}^2 \end{pmatrix}$$



III) Modelling the 3D shapes of central galaxies in their host halos

- Populating each halo with a 3D central galaxy with axis lengths A_g, B_g, C_g such that:

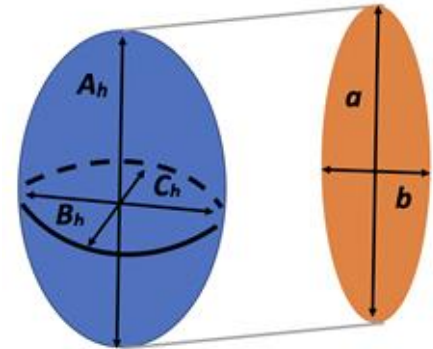
$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h} \quad \frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h} \quad \text{with } \tau_B, \tau_C \in [0; \tau_{\max}]$$

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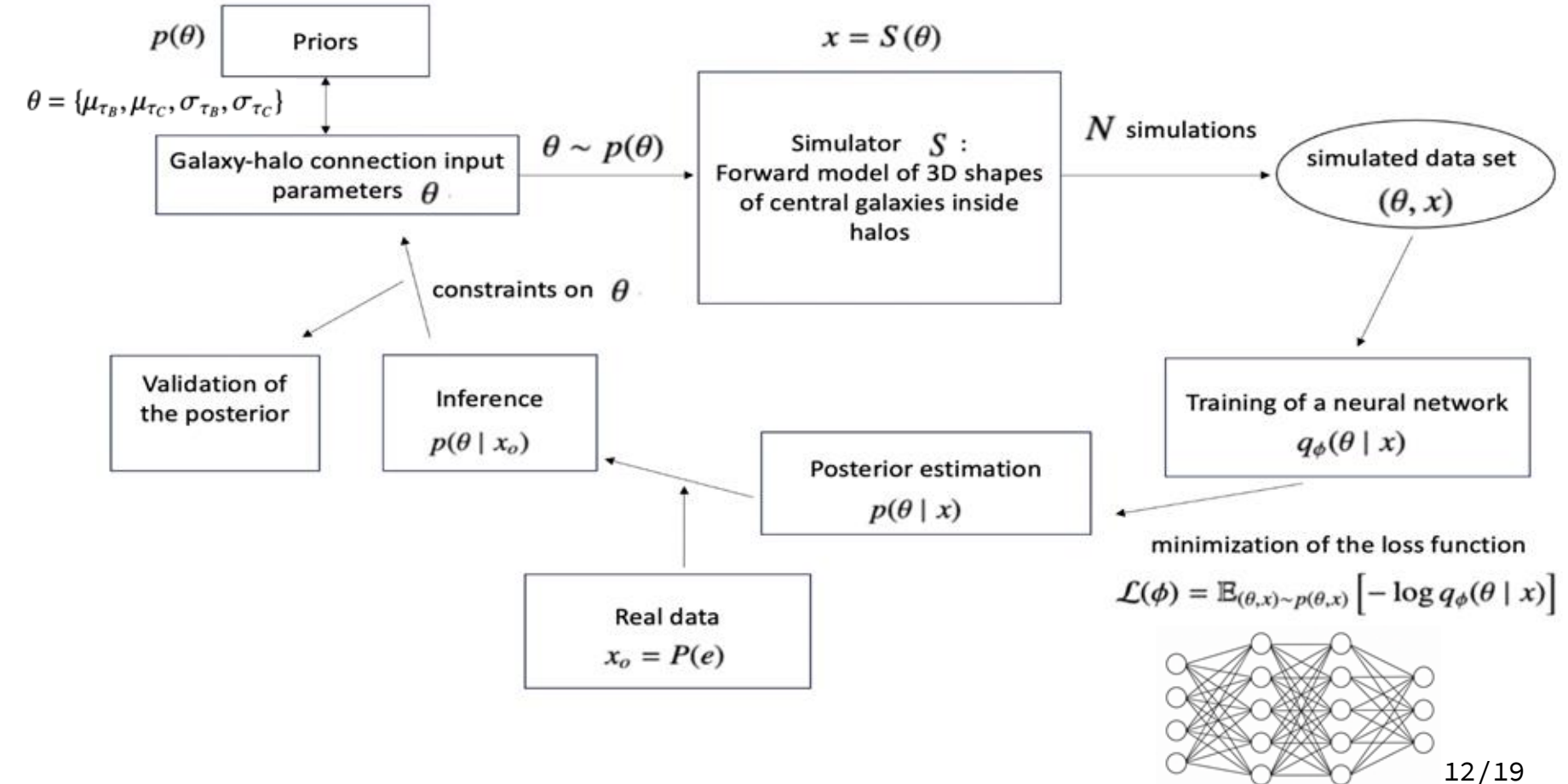
$$\Sigma = \begin{pmatrix} \sigma_{\tau_B}^2 & 0 \\ 0 & \sigma_{\tau_C}^2 \end{pmatrix}$$

- 4 free parameters: $\theta = \{\mu_{\tau_B}, \mu_{\tau_C}, \sigma_{\tau_B}, \sigma_{\tau_C}\}$
- Projection in 2D ([Lamman et al. 2023](#)) for different $\mu_{\tau_B}, \mu_{\tau_C}$ drawn with a uniform prior over $[0; 1]$.

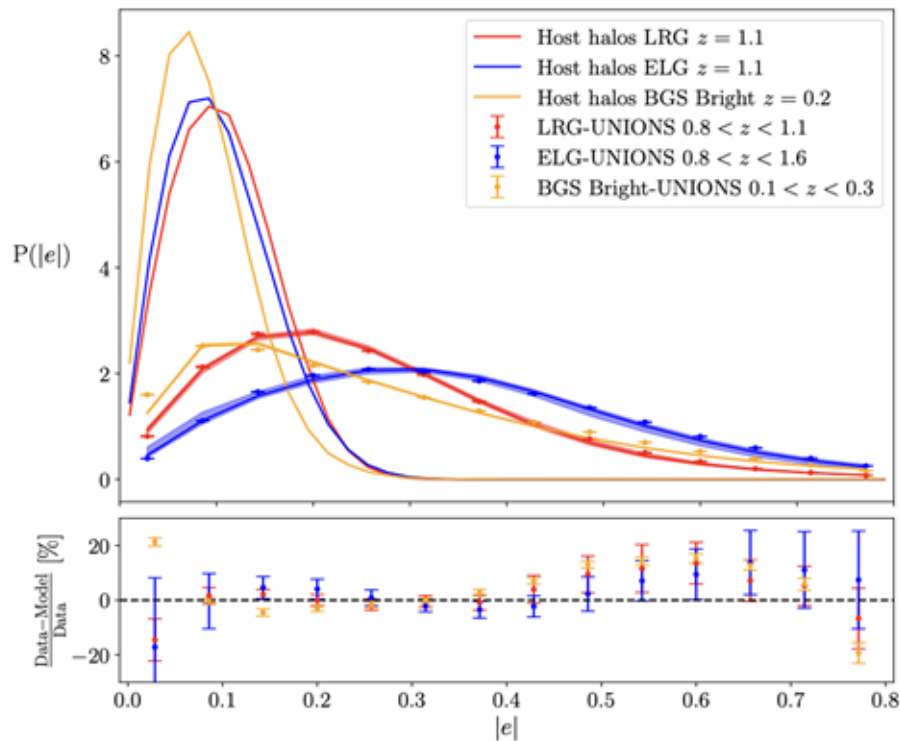
Host halo
Galaxy



IV) Inferring galaxy-halo connection with SBI



Results (1), ELG-LRG-BGS: distributions of ellipticities

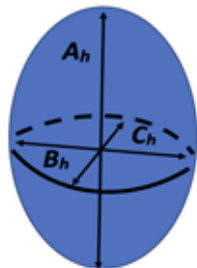


ELG 2D shapes more scattered than LRG; BGS 2D shapes rounder than ELG and LRG

Results (1), ELG-LRG-BGS: galaxy-halo connection

$\mu_{\tau_B}, \mu_{\tau_C}$: means of τ_B, τ_C
for a given halo sample

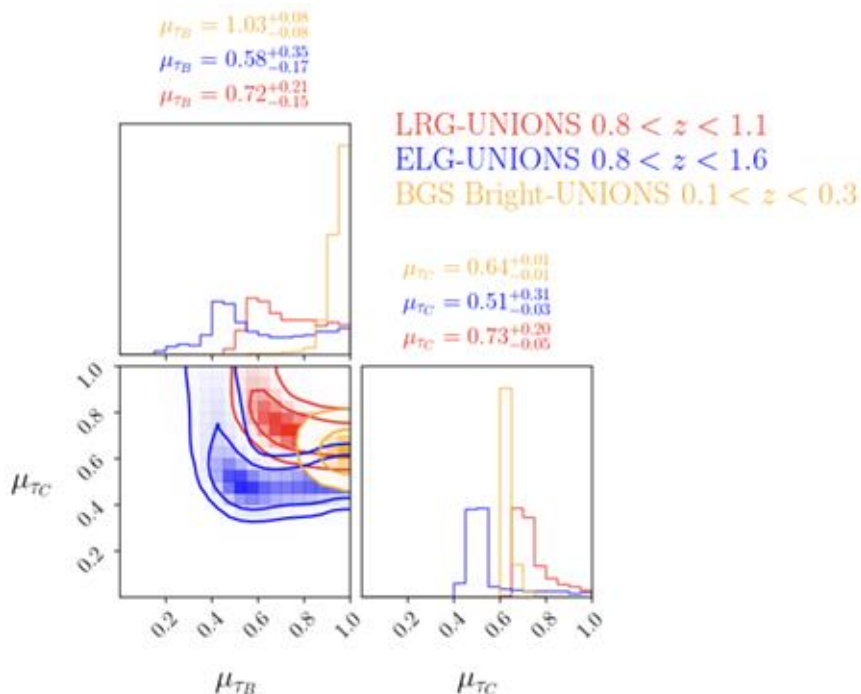
Host halo
Galaxy



$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h}$$

$$\frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h}$$

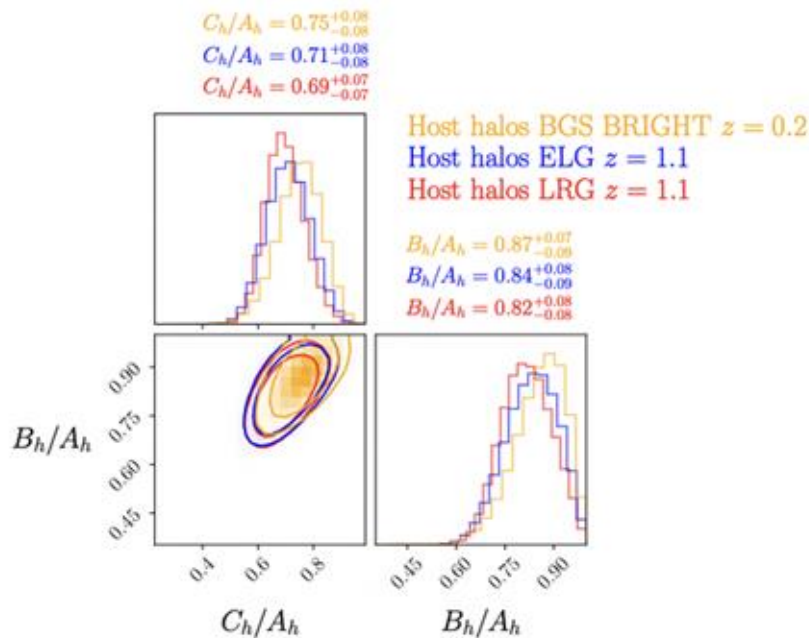
with $\tau_B, \tau_C \in [0; \tau_{\max}]$



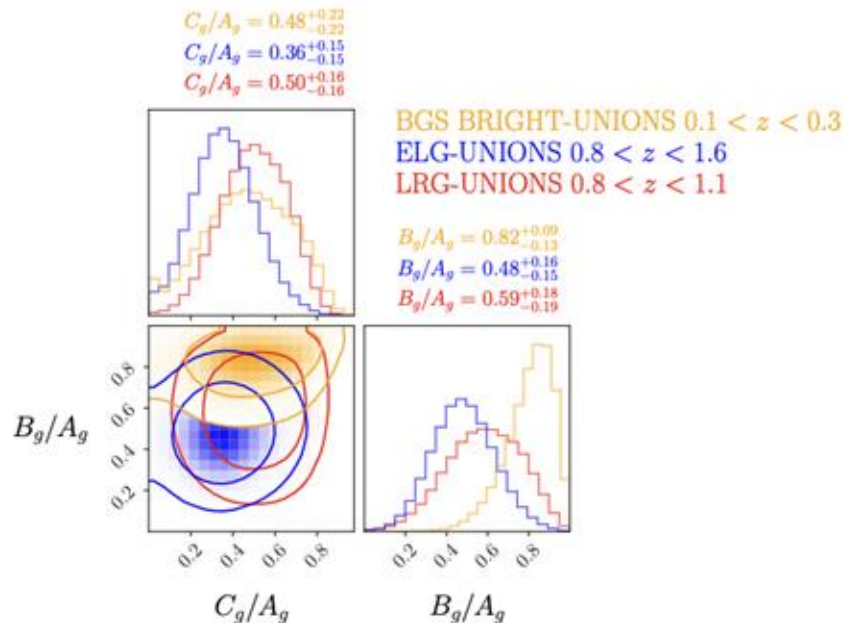
We probe different 3D galaxy-halo connections between LRG, ELG and BGS Bright

Results (1), ELG-LRG-BGS: constraints on 3D shapes

Halos

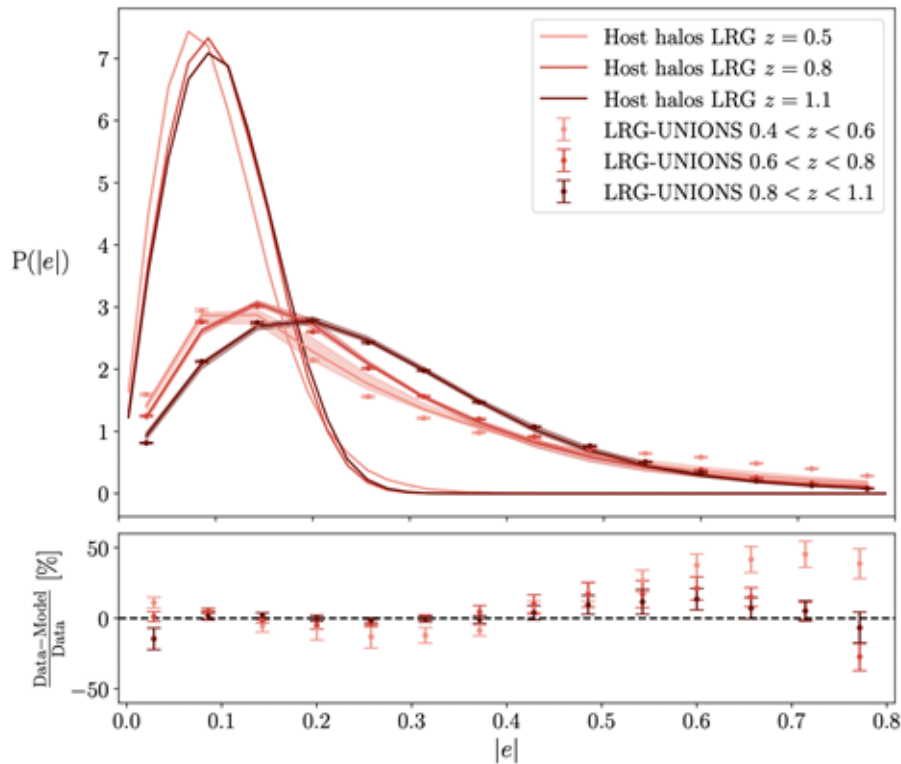


Galaxies



We estimate different 3D morphologies of galaxies depending on their types, from their observed projected images

Results (2), redshift evolution: distribution of ellipticities

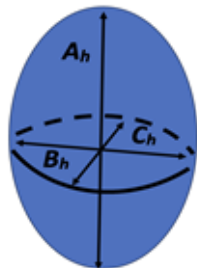


The LRG 2D shapes are more elongated at high redshift

Results (2), redshift evolution: galaxy-halo connection

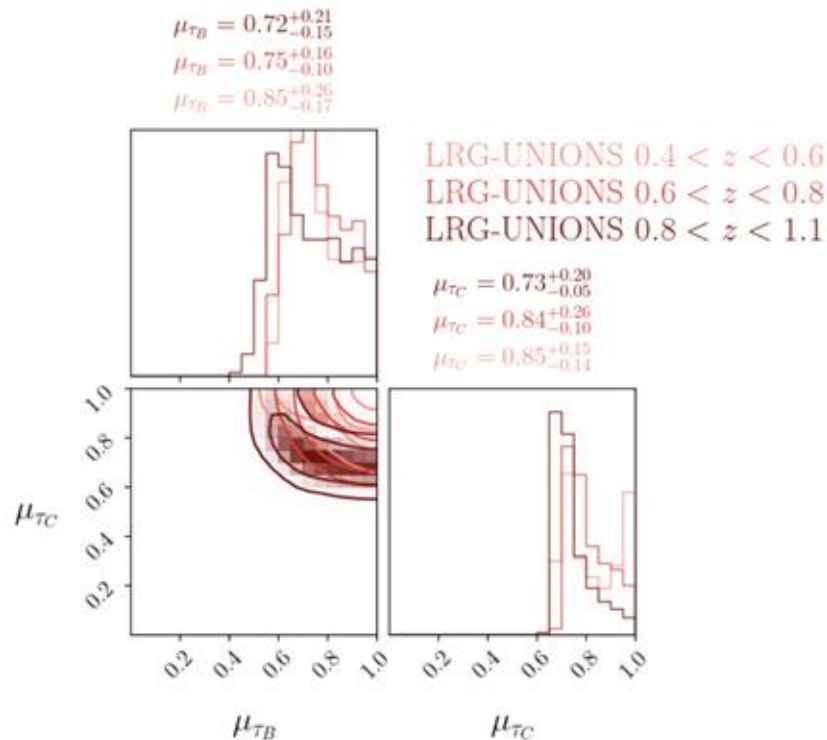
$\mu_{\tau_B}, \mu_{\tau_C}$: means of τ_B, τ_C
for a given halo sample

Host halo
Galaxy



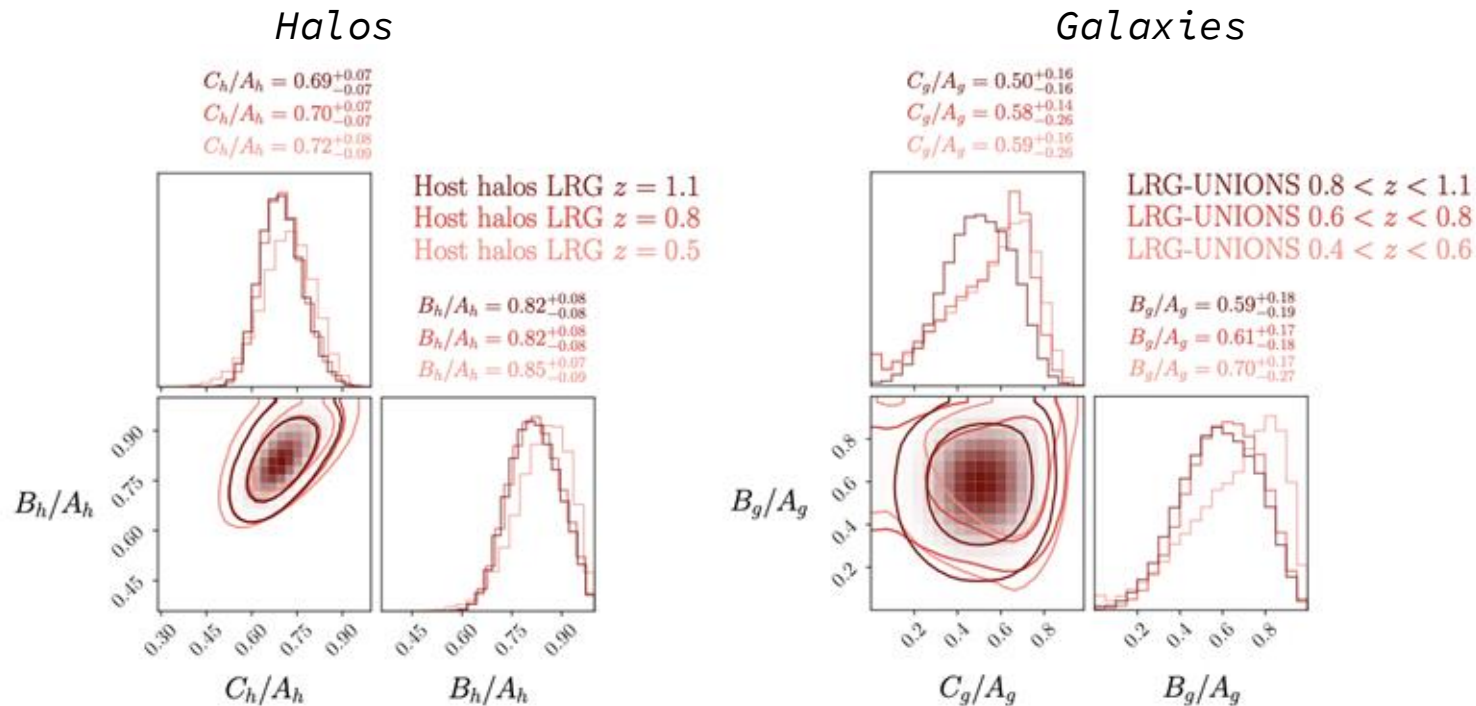
$$\frac{B_g}{A_g} = \tau_B \frac{B_h}{A_h} \quad \frac{C_g}{A_g} = \tau_C \frac{C_h}{A_h}$$

with $\tau_B, \tau_C \in [0; \tau_{\max}]$



The LRG 3D shapes are less stretched inside their halos at low redshift

Results (2), redshift evolution: constraints on 3D shapes

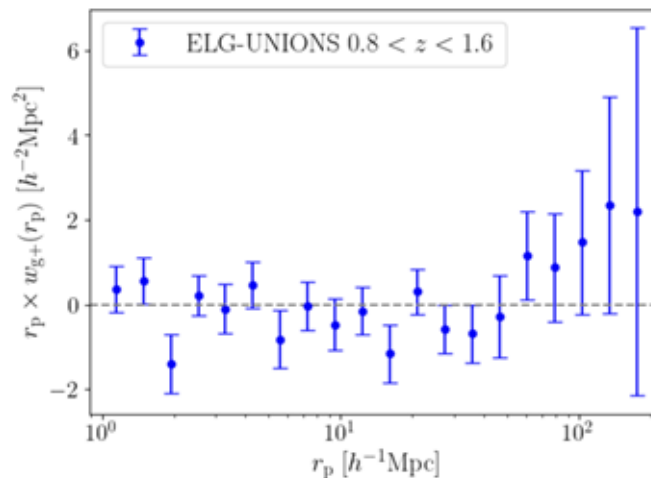
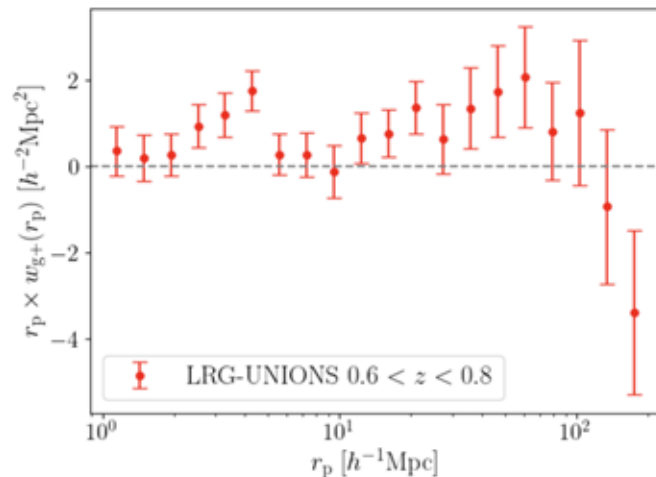


Galaxies and their host halo 3D shapes get rounder with cosmic time

Towards IA prediction with galaxy 3D morphology model

$$\xi_{g+}(r_p, \Pi) = \frac{S_+ D(r_p, \Pi) - S_+ R_D(r_p, \Pi)}{R_S R_D(r_p, \Pi)}$$

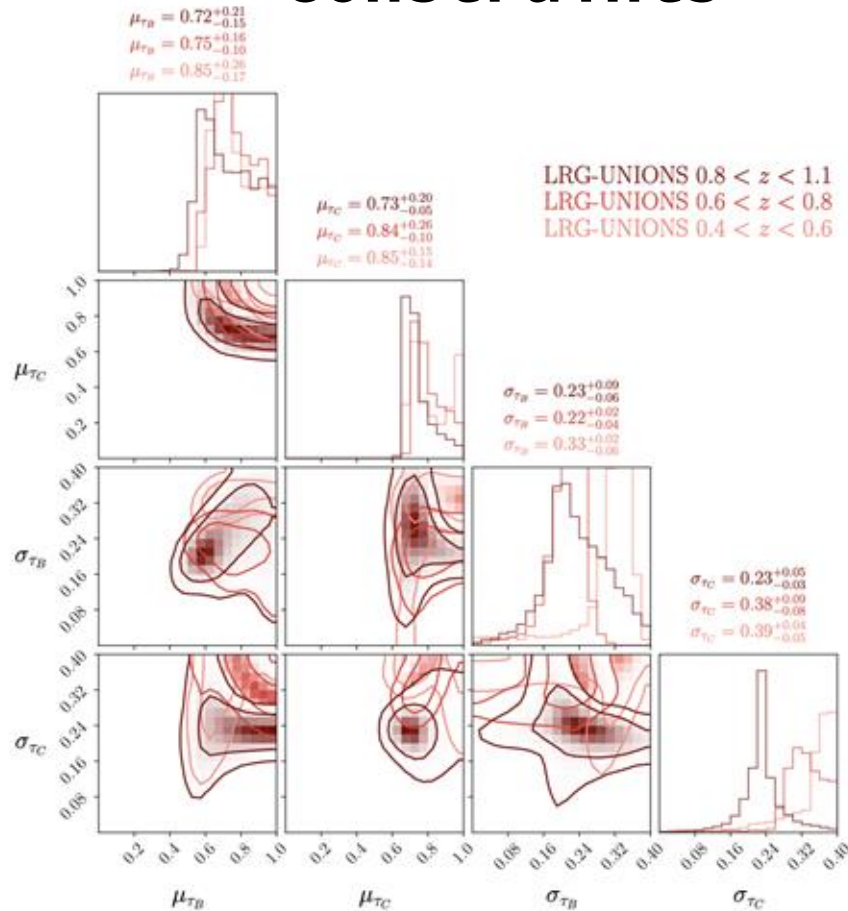
$$w_{g+}(r_p) = \int_{-\Pi_{\max}}^{\Pi_{\max}} d\Pi \xi_{g+}(r_p, \Pi)$$



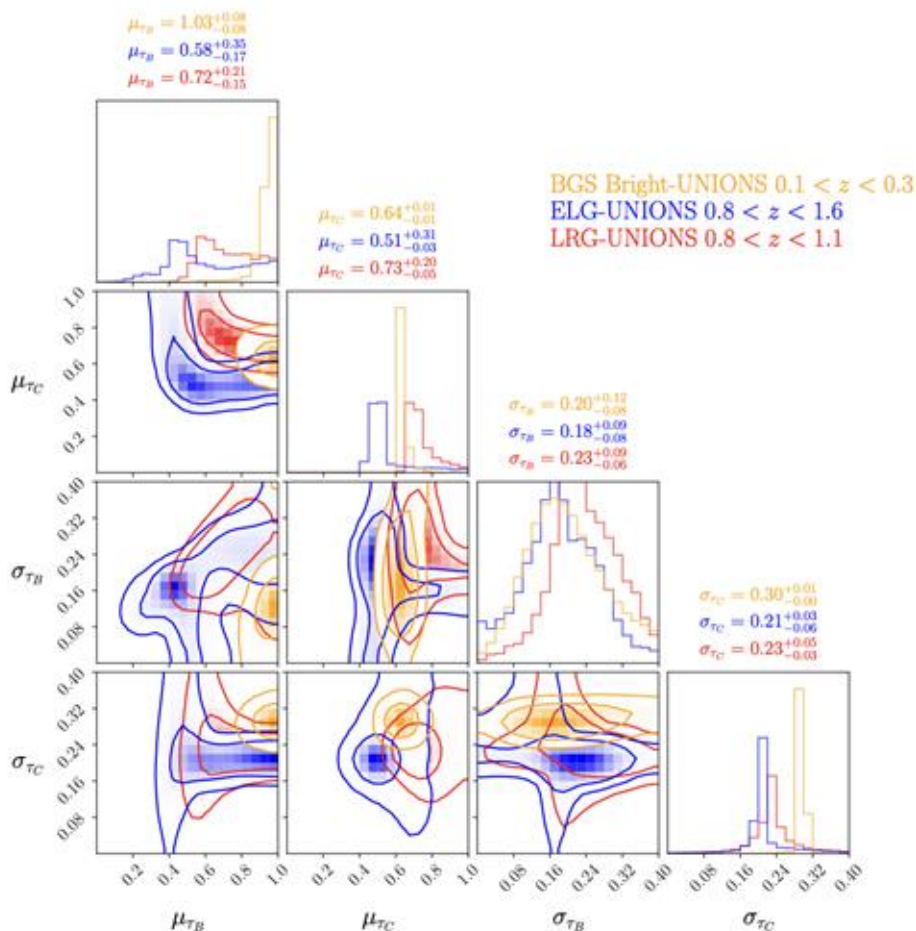
- **Future aim:** make a prediction of intrinsic alignment of galaxy using our model of 3D morphology
=> **motivation:** to reveal extra informations to intrinsic alignments about the 3D morphology of galaxies

Back up slides

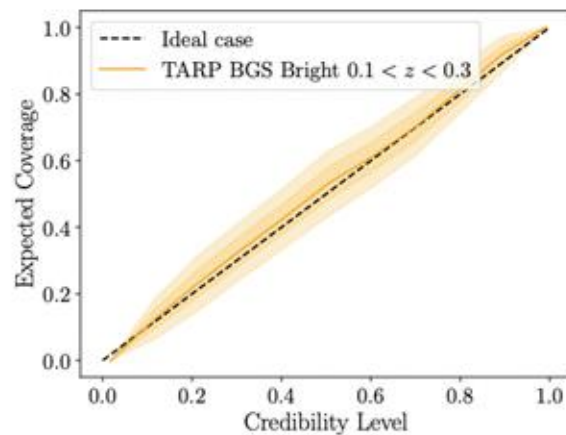
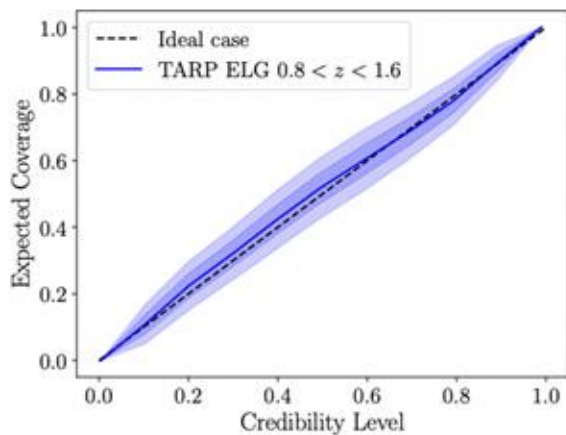
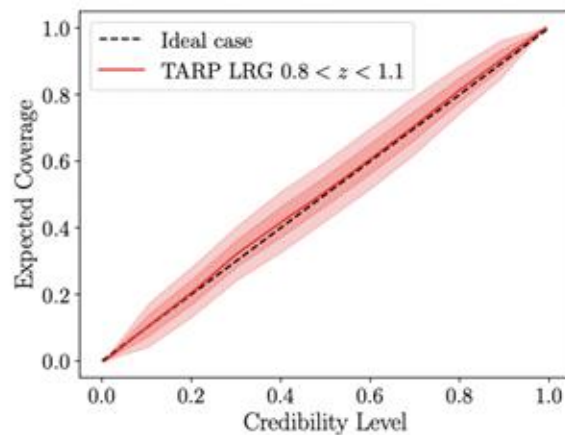
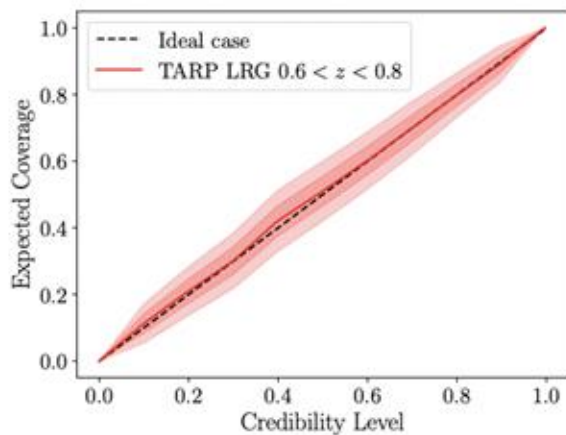
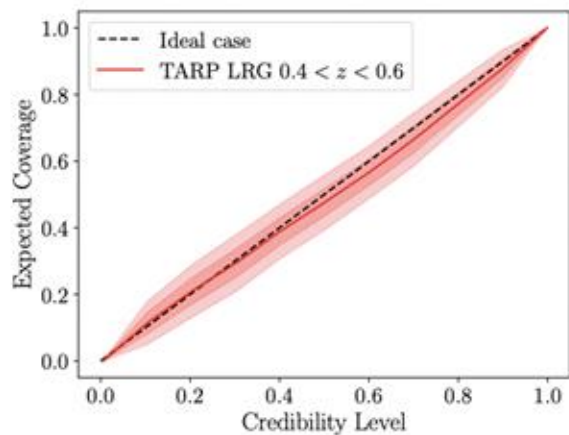
Redshift evolution: other parameter constraints



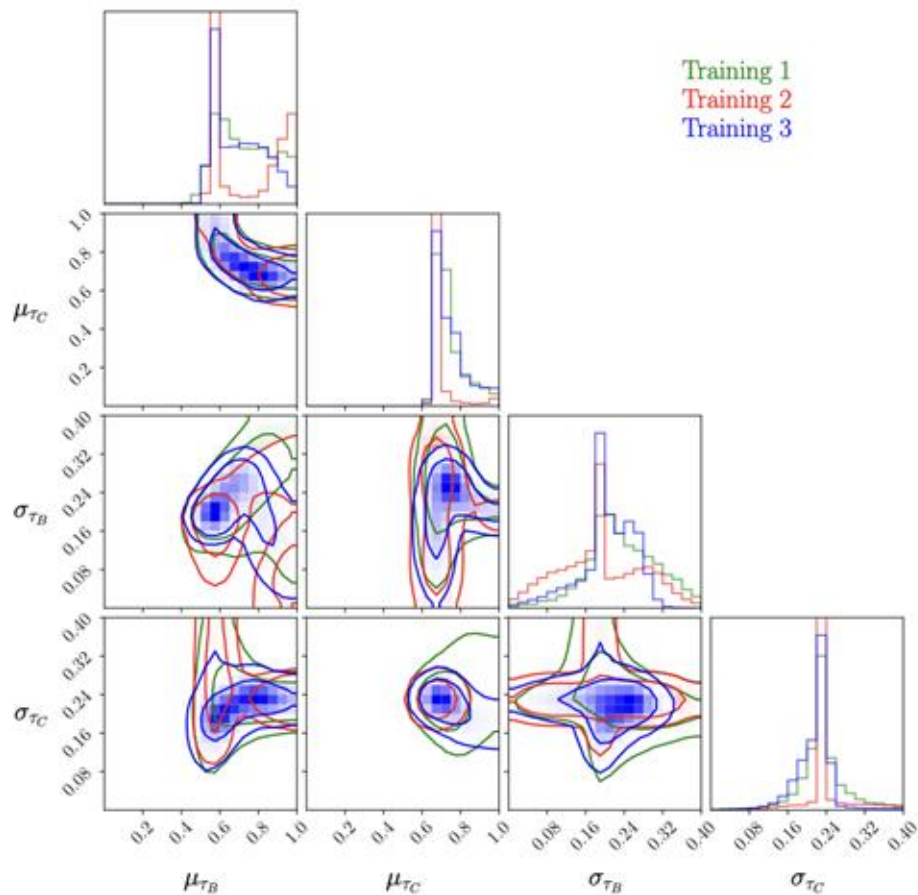
ELG-LRG-BGS: other parameter constraints



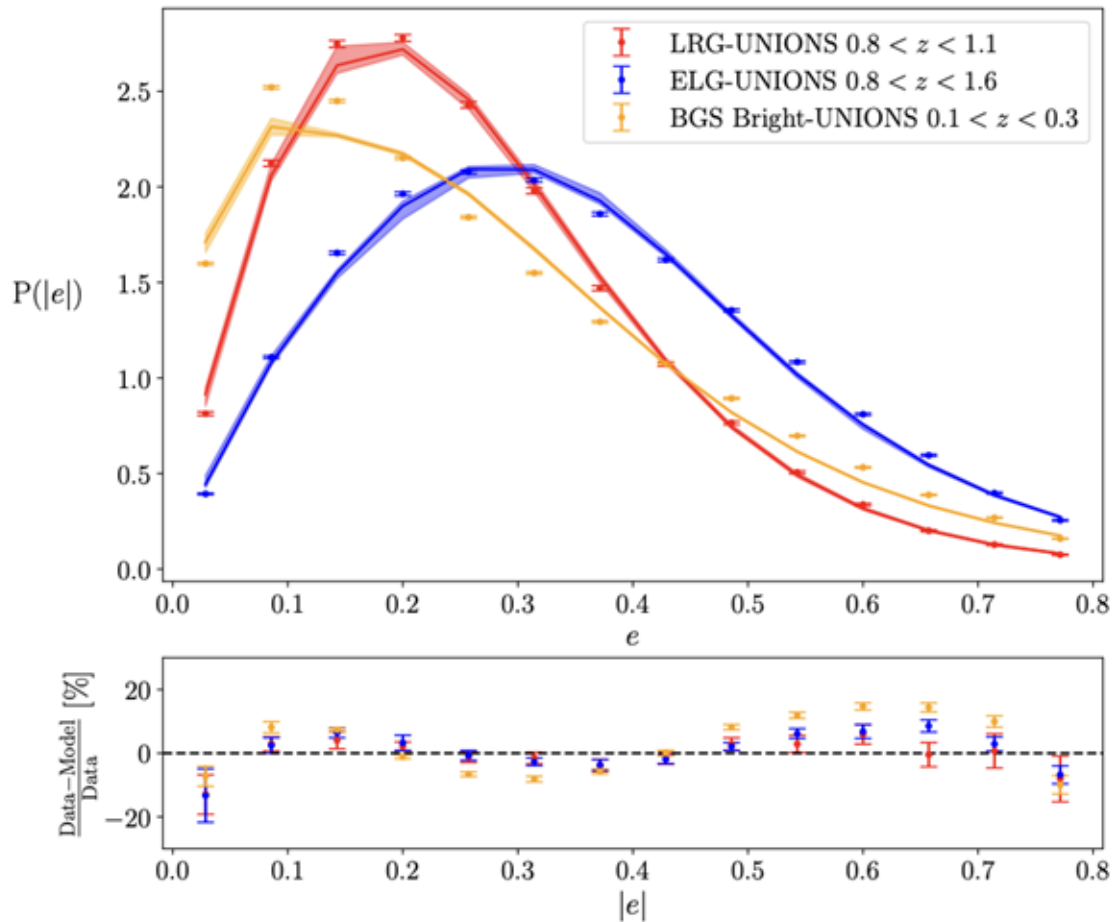
Validation: coverage tests



Validation: stability test (LRG)



Toy-model for the 3D shapes of galaxies (1)



Toy-model for the 3D shapes of galaxies (2)

